

Secure Communication Configuration

Enhancing Security in Messaging Applications

Spinnaker Analytics

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1. Introduction

1.1 Purpose

- This document defines the secure communication controls implemented to protect data transmitted between messaging clients and backend services.
- The objective is to ensure confidentiality, integrity, authentication, and forward secrecy for all communication channels used by messaging, file transfer, authentication, and voice services.

1.2 Objectives

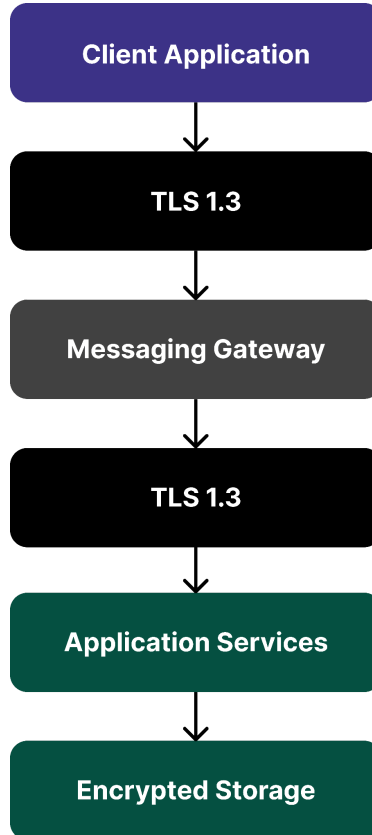
- Secure client-server communication
- Protect data in transit
- Enforce modern cryptographic standards
- Implement Perfect Forward Secrecy
- Prevent Man-in-the-Middle attacks
- Validate communication security through testing

2. Secure Communication Architecture

All communication channels are protected using layered security controls.

2.1 Secure Communication

Figure 21 – Secure Communication Architecture



2.2 Communication Channels

Channel	Protection
Authentication	TLS 1.3
Messaging	TLS 1.3 + E2EE
File Transfer	TLS 1.3 + AES-256-GCM
Voice Signaling	TLS 1.3
Voice Media	DTLS-SRTP
API Calls	TLS 1.3

3. TLS 1.3 Configuration

TLS 1.3 provides the primary transport security mechanism.

3.1 TLS Configuration

Parameter	Configuration
Protocol Version	TLS 1.3
Legacy TLS Support	Disabled
Certificate Type	X.509
Key Exchange	ECDHE
Authentication	RSA-2048 / ECDSA
Session Resumption	Enabled

HTTP Strict Transport Security (HSTS)

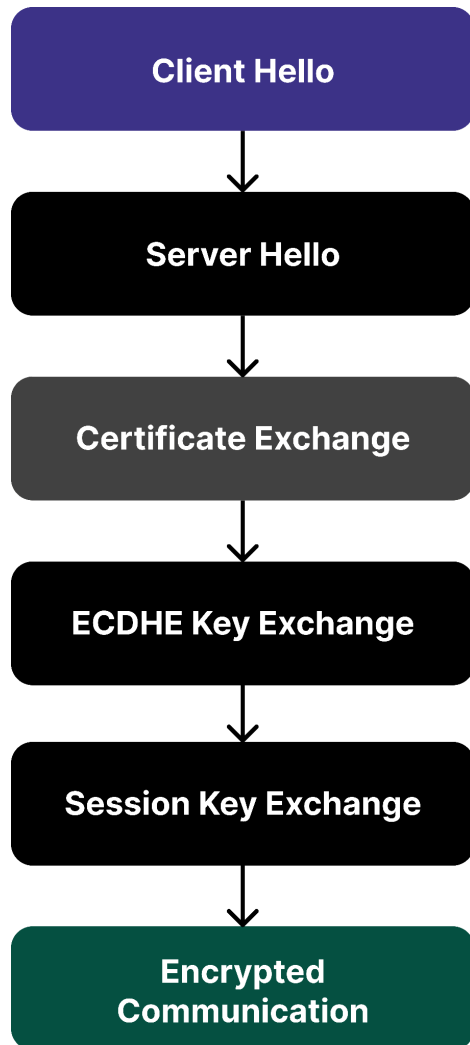
Parameter	Configuration
Enabled	Yes
Max Age	31536000
Include Subdomains	Yes

3.2 Supported Protocols

Protocol	Status
SSL 2.0	Disabled
SSL 3.0	Disabled
TLS 1.0	Disabled
TLS 1.1	Disabled
TLS 1.2	Optional
TLS 1.3	Enabled

3.3 TLS Handshake

Figure 22 – TLS Handshake Process



4. Certificate Management

Certificates are used to authenticate servers and establish trusted communication.

4.1 Certificate Requirements

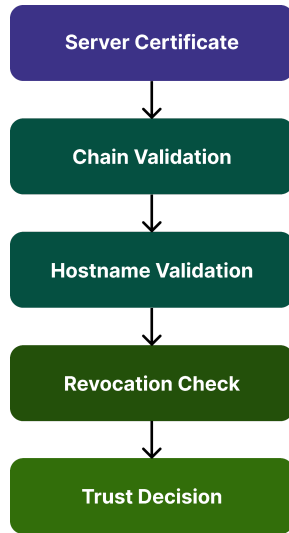
Parameter	Value
Format	X.509
Key Size	RSA-2048 Minimum
Signature Algorithm	SHA-256
Validity Period	365 Days
Revocation Support	CRL + OCSP

4.2 Certificate Validation Controls

- Validation Checks
 - Certificate chain validation
 - Expiration validation
 - Hostname verification
 - Revocation checking

4.3 Certificate Validation

Figure 23 – Certificate Validation Workflow



4.4 Certificate Pinning

- Certificate pinning is implemented within mobile clients to prevent acceptance of fraudulent certificates even if a trusted Certificate Authority is compromised.
- **Benefits**
 - Protection against rogue certificates
 - Mitigation of MITM attacks
 - Enhanced trust validation
- **Pinning Method**

Parameter	Value
Pin Type	Public Key Pinning
Validation	SHA-256 Hash
Backup Pins	Supported

5. Perfect Forward Secrecy Implementation

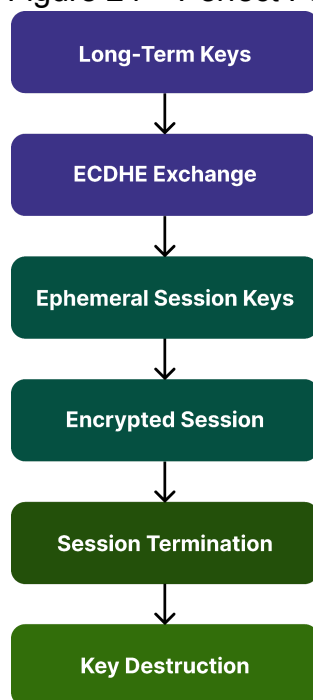
Perfect Forward Secrecy ensures that historical communications remain protected even if long-term keys are compromised.

5.1 Technology Used

Technology	Purpose
ECDHE	Ephemeral Key Exchange
X3DH	Session Establishment
Double Ratchet	Message Key Rotation
DTLS	Voice Session Security

5.2 Perfect Forward Secrecy

Figure 24 – Perfect Forward Secrecy Workflow



5.3 Benefits

- Historical message protection
- Session isolation
- Reduced compromise impact
- Secure key rotation

6. DTLS Configuration

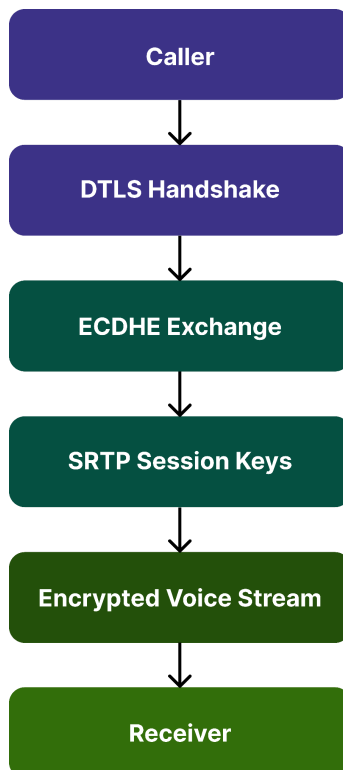
- Datagram Transport Layer Security protects voice communication channels.

6.1 DTLS Settings

Parameter	Configuration
Protocol	DTLS 1.2
Key Exchange	ECDHE
Cipher	AES-128-GCM
Authentication	DTLS Certificates
PFS Support	Enabled

6.2 DTLS-SRTP Architecture

Figure 25 – DTLS-SRTP Architecture



7. Cipher Suite Configuration

Only modern cryptographic suites are permitted.

7.1 Approved TLS Cipher Suites

Cipher Suite	Status
TLS_AES_256_GCM_SHA384	Enabled
TLS_CHACHA20_POLY1305_SHA256	Enabled
TLS_AES_128_GCM_SHA256	Enabled

7.2 Deprecated Cipher Suites

Cipher Suite	Status
RC4	Disabled
DES	Disabled
3DES	Disabled
MD5-Based Suites	Disabled

7.3 Security Controls

- Strong encryption only
- Authenticated encryption
- Forward secrecy support
- Secure key exchange

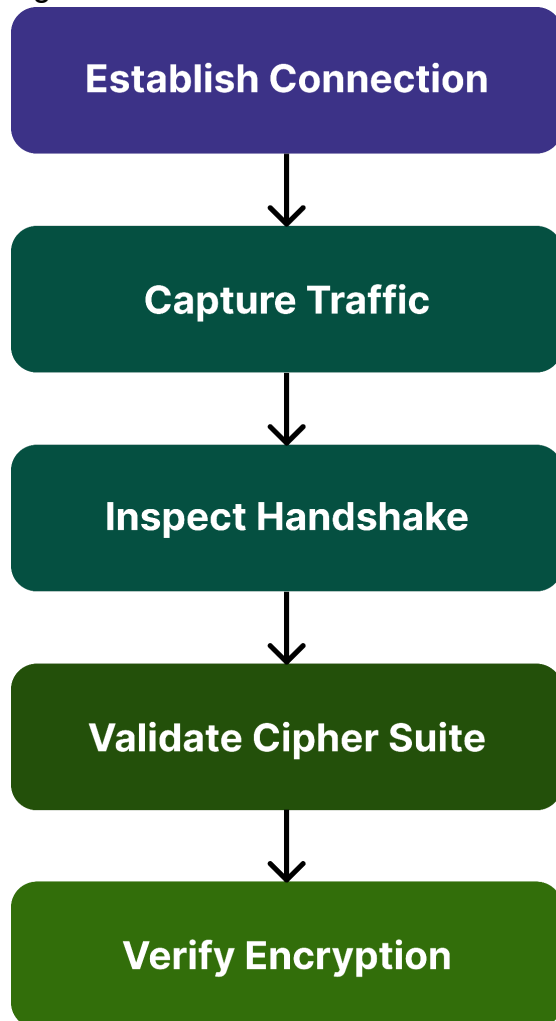
8. Communication Security Validation

Communication controls must be validated through testing.

8.1 Testing Environment

Tool	Purpose
Wireshark	Packet Analysis
Burp Suite Community	TLS Verification
OpenSSL	Certificate Testing
SSL Labs	TLS Assessment

Figure 26 – Validation Process



8.2 Validation Objectives

- **TLS Validation**
 - Verify:
 - TLS 1.3 active
 - Certificate validation
 - Secure cipher suites
- **DTLS Validation**
 - Verify:
 - DTLS handshake
 - SRTP key generation
 - Encrypted voice traffic

8.3 Test Cases

Test ID	Test Scenario	Expected Result
COMM-01	TLS Connection	TLS 1.3 Established
COMM-02	Certificate Validation	Trusted Certificate
COMM-03	Cipher Inspection	Approved Cipher
COMM-04	DTLS Handshake	Successful
COMM-05	Voice Session	SRTP Active
COMM-06	MITM Simulation	Connection Blocked

8.4 Expected Evidence

- Wireshark captures
- TLS handshake screenshots
- OpenSSL verification results
- SSL Labs assessment report

Validation Area	Expected Outcome
TLS Version	TLS 1.3
Certificate Chain	Trusted
Cipher Suite	Approved
PFS	Verified
DTLS Session	Successful
SRTP Traffic	Encrypted

9. Security Considerations

9.1 Threats & Mitigations

Threat	Mitigation
MITM Attacks	TLS 1.3
Certificate Spoofing	Certificate Validation
Weak Cipher Attacks	Approved Cipher Suites
Session Hijacking	ECDHE
Replay Attacks	Nonces
Downgrade Attacks	TLS 1.3 Enforcement

10. Standards Alignment

Standard	Purpose
RFC 8446	TLS 1.3
RFC 6347	DTLS
RFC 5246	TLS Framework
RFC 5280	X.509 Certificates
NIST SP 800-52	TLS Guidance
OWASP MASVS	Mobile Security

11. Conclusion

- This Secure Communication Configuration document defines the controls used to secure all communication channels within the messaging application.
 - The architecture combines:
 - TLS 1.3 transport security
 - DTLS-secured voice communication
 - Strong certificate management
 - Modern cipher suites
 - Perfect Forward Secrecy
 - Comprehensive communication validation
- These controls ensure secure transmission of messages, files, authentication data, and voice communications while protecting against interception, spoofing, downgrade attacks, and unauthorized access.